DATA ITEM DESCRIPTION

Title: CRITICAL SIGNAL ANALYSIS

Number: DI-NUOR-82444
AMSC Number: F10477
DTIC Applicable: No
Preparing Activity: 37 (AFSEC/SEWN)

Approval Date: 20240322
Limitation: N/A
GIDEP Applicable: No
Project Number: NUOR-2023-005

Applicable Forms: N/A

Use/relationship: A Critical Signal Analysis (CSA) describes the flow of all Critical Signals throughout a nuclear weapon system and identifies the components involved in the initiation and propagation of Critical Signals. Critical Signals are the signals that control or contribute to the activation and deactivation of Critical Functions within a nuclear weapon. The System Program Office generates and maintains the CSA and presents it to Headquarters Air Force Safety Center Nuclear Weapons Safety Branch (AFSEC/SEWN) as part of Nuclear Safety Design Certification (NSDC). The CSA is a living document and will be modified as necessary throughout the product's lifecycle continually and accurately to describe the nuclear weapon system's Critical Signals.

Chapter 3 of the Department of the Air Force Manual (DAFMAN) 91-119, Safety Design and Evaluation Criteria for Nuclear Weapon Systems Software describes the CSA. DAFMAN 91-119 and its requirements implement Department of the Air Force Instruction (DAFI) 91-101, Air Force Nuclear Surety Program, as part of the NSDC process. The CSA is required as supporting documentation for any Nuclear Certification Impact Statement (NCIS) and Nuclear Surety Evaluation (NSE) that meets the criteria chapter 3 of DAFMAN 91-119 names.

a. This Data Item Description (DID) contains the format, content, and intended use information for the data deliverable resulting from the work task described in the solicitation and defined in Chapter 3 of DAFMAN 91-119 and is applicable to the acquisition of software for Department of the Air Force systems requiring NSDC.

b. DAFMAN 91-119 defines information required to implement this DID. This information includes the guidelines for content to be contained in a CSA and when a CSA is to be created and provided to AFSEC/SEWN.

Requirements:

1. **Reference documents.** The applicable issue of the documents cited herein, including their approval dates and the dates of any applicable amendments, notices, and revisions, shall be

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as specified in the contract.

2. Format. The CSA shall be in the contractor's format.

2.1 The CSA is not limited in length, but it is limited in scope by the requirements laid out in section 3 of this DID.

2.2 The CSA shall be submitted to AFSEC/SEWN as a static product that is viewable locally in its entirety with all referenced source documentation submitted in conjunction with the CSA.

3. Content. The CSA shall contain an analysis of all Critical Signals within the nuclear weapon system.

3.1 The CSA shall contain a description of the nuclear weapon system as it pertains to Critical Signals for the purpose of NSDC. This includes test equipment that is required to undergo the NSDC process per DAFI 91-101.

3.2 If the item or system undergoing NSDC is a modification of an existing item or system, the CSA shall contain a description of the modification as it pertains to the Critical Signals within the nuclear weapon system

3.3 The CSA shall contain a description of the flow of each Critical Signal pertaining to the nuclear weapon system, tracing from human intent to Critical Function activation, and including all system prerequisites pertaining to the initiation and propagation of all Critical Signals. Figure 1and Figure 2 are examples of generic system level diagrams depicting traces of Critical Signals through a nuclear weapon system.

3.3.1 The CSA shall show all components along the path of the Critical Signal.

3.3.1.1 The CSA shall identify all components along the path of the Critical Signal regardless of the certification status of the component.

3.3.1.2 The CSA shall identify any components used for the navigation or guidance of a nuclear weapon for the purpose of performing Navigation Validation. (Inertial Navigation Units, gyroscopes, accelerometers, etc.)

3.3.1.3 The CSA shall contain a description of the mechanisms by which the nuclear weapon system operators provide human intent for the initiation or propagation of Critical Signals such as keyboards, display bezel presses, switches, etc.

3.3.1.4 The CSA shall contain a description of all components along the path of the Critical Signal that are responsible for changing the format or protocol of the Critical Signal for the purpose of further propagation within the nuclear weapon system.

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3.3.2 The CSA shall provide an analysis of the potential effects all components along the path of the Critical Signals have on the Critical Signals.

3.4 The CSA shall describe the mechanisms used to protect the Critical Signals within the nuclear weapon system such as encryption, digital signatures, hashes, etc.

3.5 The CSA shall describe the methods by which software components involved in the creation or propagation of Critical Signals are protected from other unauthorized software components or inadvertent software interactions. These methods can include partitioning, memory management, or other forms of protection.

3.6 In the case of test equipment requiring NSDC, the CSA shall meet the following sub-requirements:

3.6.1 The CSA shall identify any components under test that are responsible for the initiation or propagation of Critical Signals, or any components responsible for the propagation of Safing Commands.

3.6.2 The CSA shall contain a description of all test events that pertain to the verification of operation of components that initiate or propagate Critical Signals throughout the weapon system, or that pertain to the verification of operation of components for making the nuclear weapon system safe.

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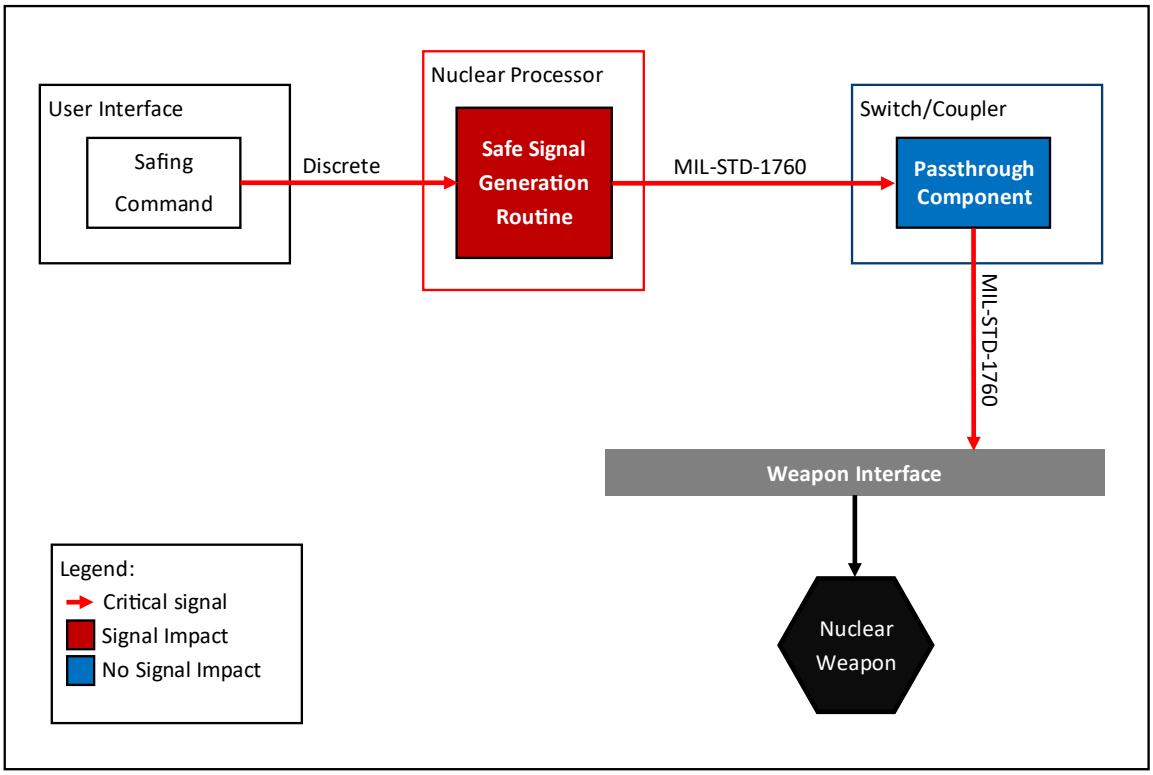


Figure 1. Example Critical Signal trace – Safing Command

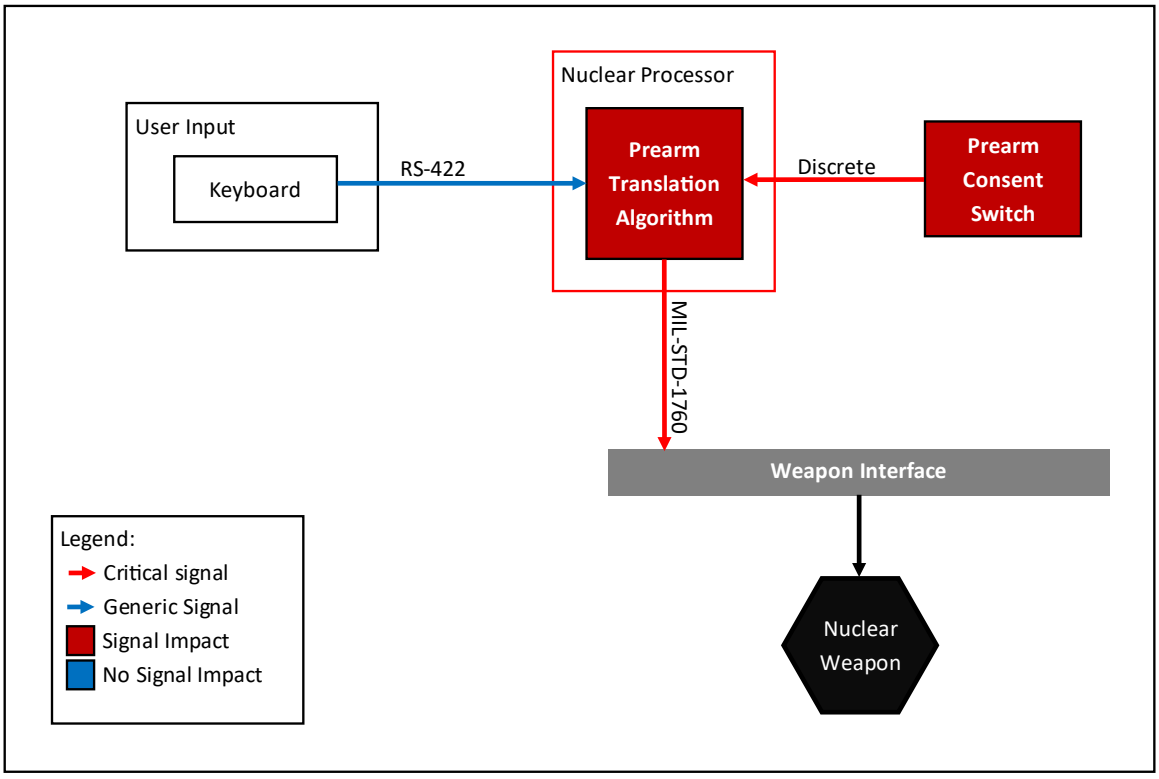


Figure 2. Example Critical Signal trace – Prearm

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